COMPLEMENTS OF REEB-INVARIANT IMMATERIAL SETS SURJECT ONTO SYMPLECTIC HOMOLOGY

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Allowed standard prerequisites (editorial).

- Standard contact/Floer continuation and naturality. Original complete bibliography and source-local definitions retained.

A subset A of a cooriented contact manifold M is called *immaterial*[1] if there exists a contractible loop of contactomorphisms $\varphi_t\colon M \to M$ whose vector field X points in the positive direction (of the contact distribution) on A. The property of being immaterial is clearly related to orderability, a well studied phenomenon in contact geometry originally introduced in [6]. Namely, a contact manifold M is non-orderable if, and only if, all of its subsets are immaterial. The existence of non-orderable contact manifolds shows that immaterial subsets are not necessarily small in the sense of contact squeezing or contact displaceability. Indeed, the whole contact manifold M (within itself) cannot be squeezed into a smaller subset or displaced from itself, although the whole M is immaterial if non-orderable. The only immaterial subset in dimension 1 is the empty set.[2] This is in contrast with dimensions greater than 3 where every compact subset of a Darboux chart is immaterial [19, Lemma 7.3].

The results of this note can be seen as contributions to questions regarding the size in contact geometry. These questions are quite delicate (as evident from work by Eliashberg–Kim–Polterovich [5]) and have been an active research area over the last two decades. Our first result (Theorem 1) is formulated in terms of selective symplectic homology [19], while the second (Theorem 2) is expressed using recently introduced contact quasi-measures [4]. The contact quasi-measures from [4] are strongly inspired by Entov–Polterovich's construction of (symplectic) quasi-measures [8] using spectral invariants from [13]. See [7] or [14] for a more detailed historical account.

A central role in this paper is played by *contact Hamiltonian Floer homology*, introduced in [12] by proving a maximum principle for slopes that are allowed to be general contact Hamiltonians (instead of constants in $\mathbb{R}$ as common in symplectic homology theory). The cases where the contact Hamiltonian (i.e., the slope) is positive [17] or Reeb-invariant [15] were known prior to [12]. The theory of contact Hamiltonian Floer homology was further developed in [3, 20] and in [1, 2] with independent contributions.

0.1. **Immaterial subsets and symplectic homology.** In this section, W denotes a Liouville domain.[3] Selective symplectic homology $SH^{\Omega}_{*}(W)$ is a version

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[1]The notion was introduced in [19, Section 7].

[2]Here, we prove the claim for $\mathbb{S}^1$. It is enough to show that a point $p \in \mathbb{S}^1$ is not immaterial. Let $\varphi_t\colon \mathbb{S}^1 \to \mathbb{S}^1$ be a loop of contactomorphisms that is positive at p. Then, the degree of the map $\gamma\colon \mathbb{S}^1 \to \mathbb{S}^1$, $t \mapsto \varphi_t(p)$ is positive. (This can be seen by counting elements of $\gamma^{-1}(\{p\})$ with signs.) Therefore, $\{\varphi_t\}_t$ cannot be contractible.

[3]We expect the result of this section to hold in a more general framework than that of Liouville domains. See [19, Remark 1.5].

of symplectic homology introduced in [19]. It is obtained as the direct limit of Floer homology groups for Hamiltonians whose slopes tend to $+\infty$ on the conical end above the open subset $\Omega \subset \partial W$ of the boundary, but remain close to 0 and positive elsewhere. The more precise definition of $SH_*^{\Omega}(W)$ is recalled in Section 1. If $\Omega = \partial W$, then $SH_*^{\Omega}(W)$ coincides with Viterbo's [21] symplectic homology $SH_*(W)$. Given two open subsets $\Omega_1 \subset \Omega_2$ of ∂W, there is a functorial morphism $SH_*^{\Omega_1}(W) \to SH_*^{\Omega_2}(W)$ called *continuation map*. In particular, the continuation map $SH_*^{\Omega}(W) \to SH_*(W)$ is well defined for every open subset $\Omega \subset \partial W$. The following theorem shows that complements of immaterial Reeb-invariant subsets are big in terms of selective symplectic homology.

Theorem 1. *Let $A \subset \partial W$ be the preimage of a closed subset under a smooth Reeb-invariant map $\partial W \to \mathbb{R}$. Assume A is immaterial and denote $\Omega := \partial W \setminus A$. Then, the continuation map*

$$SH_*^{\Omega}(W) \to SH_*(W)$$

is surjective.

Theorem 1 can be seen as a generalization and an improvement of [19, Theorem 7.4]. Namely, [19, Theorem 7.4] claims that the rank of the map $SH_*^{\Omega}(W) \to SH_*(W)$ is equal to $\dim SH_*(W)$ in the case where A is a transversely embedded circle. It is an interesting question whether the condition of A being the preimage of a closed subset under a smooth Reeb-invariant map $\partial W \to \mathbb{R}$ can be relaxed.

0.2. **Contact quasi-measures and immaterial subsets.** In this section, W stands for a weakly$^+$ monotone strong filling of a closed contact manifold such that the unit $e \in SH_*(W)$ is not eternal. In what follows, we recall the notions of a weakly$^+$ monotone symplectic manifold and of an eternal element. A $2n$-dimensional symplectic manifold (W, ω) is weakly$^+$ monotone[4] if at least one of the following conditions is satisfied:

(1) $\omega|_{\pi_2(W)} = 0$,
(2) $c_1|_{\pi_2(W)} = 0$,
(3) there exists $s > 0$ such that[5] $\omega|_{\pi_2(W)} = s \cdot c_1|_{\pi_2(W)}$,
(4) the minimal Chern number of W is at least $n - 1$.

These conditions ensure that the symplectic homology of W is well defined. Now, we also recall the notion of being eternal [2]. An element $\theta \in SH_*(W)$ is called *eternal* if it is contained in the image of the canonical morphism

$$HF_*(H^a) \to SH_*(W)$$

for all admissible[6] $a \in \mathbb{R}$. Here $H_t^a \colon \widehat{W} \to \mathbb{R}$ denotes a non-degenerate Hamiltonian on the completion $\widehat{W}$ of W having slope a at the conical end.

Inspired by work of Entov and Polterovich [7, 8, 9, 10], the paper [4] introduced a map τ, called *partial contact quasi-measure*, which assigns a number from $[0, 1]$ to a closed subset of ∂W. The quasi-measure τ is constructed using contact spectral invariants $c(h, \theta)$ introduced independently and simultaneously in [1] and [3]. We recall the construction of τ in Section 2. The following theorem asserts that each contractible loop of contactomorphisms furnishes a lower bound for τ.

Theorem 2. *Let W and τ be as above, and $B \subset \partial W$ a closed subset. Let $f \colon \partial W \times [0, 1] \to \mathbb{R}$ be a contact Hamiltonian giving rise to a contractible loop*

[4]See [11, Lemma 1.1], [16, Assumption (W^+)], and [15, Section 2B].

[5]Here, both ω and c_1 are seen as maps assigning a number to an element of $\pi_2(W)$.

[6]Recall that a number $a \in \mathbb{R}$ is *admissible* if it is not the period of a closed Reeb orbit on ∂W. The extension of this notion to contact Hamiltonians is recalled on page 5.

of contactomorphisms. Denote by m and m_B the infima of f over $\partial W \times [0,1]$ and $(\partial W \setminus B) \times [0,1]$, respectively. Then, the following inequality holds:

$$\tau(B) \geqslant \frac{m_B}{m_B - m}$$

if $m_B \neq m$.

By definition, $\tau(B)$ is an element of $[0,1]$. In this paragraph, we explain why Theorem 2 does not contradict this fact. The number m cannot be positive, because that would imply that ∂W is not orderable and contradict [4, Theorem 1.6]. Therefore, $m_B - m \geqslant m_B$ and $\frac{m_B}{m_B - m} \leqslant 1$. This means that Theorem 2 never forces $\tau(B)$ to be greater than 1.

It is useful to observe that Theorem 2 is only interesting in the case where $\partial W \setminus B$ is immaterial. Indeed, if m_B is negative, then the theorem asserts that $\tau(B)$ is greater than some negative number, and that follows directly from the definition of τ. Lemma 9.3 from [4] shows $\tau(B) = 0$ if B is displaceable and the preimage of a closed subset under a Reeb-invariant function. Therefore, Theorem 2 and [4, Lemma 9.3] imply the following corollary.

Corollary 3. *Let $B \subset \partial W$ be the preimage of a closed subset under a Reeb-invariant smooth function $\partial W \to \mathbb{R}$. If $\overline{\partial W \setminus B}$ is immaterial, then B is contact non-displaceable.*

It is an interesting question whether the condition of B being the preimage of a closed subset under a smooth Reeb-invariant map $\partial W \to \mathbb{R}$ can be relaxed. The next example applies Theorem 2 in a concrete situation of $S^*\mathbb{S}^n$. The example illustrates a general method of finding immaterial subsets in unit cotangent bundles by lifting contractible loops of diffeomorphisms from the base.

Example 4. Let $n \in \mathbb{N}$ be a number equal to 3 modulo 4 and let $\psi_t \colon \mathbb{S}^n \to \mathbb{S}^n$ be given by $\psi_t(x) := \mathrm{e}^{2\pi \mathrm{i} t} \cdot x$ (we see $\mathbb{S}^n$ as the unit sphere in the complex space). Since $n \geqslant 3$ is equal to 3 modulo 4, ψ_t is a contractible loop of diffeomorphisms.[7] Therefore,

$$\varphi_t \colon \ S^*\mathbb{S}^n \to S^*\mathbb{S}^n, \qquad v^* \mapsto \psi_t^*(v^*)$$

is a contractible loop of contactomorphisms. The contact Hamiltonian of φ_t is given by

$$f \colon \ S^*\mathbb{S}^n \to [-2\pi, 2\pi], \qquad f(v^*) := v^*(X(p)),$$

where X is the vector field of ψ_t and p is the base point for v^*. Denote by B_c the subset $\{v^* \in S^*\mathbb{S}^n \mid v^*(X) \leqslant c\}$. The set B_c for $c \in (-2\pi, 2\pi)$, can be seen as a fibration over $\mathbb{S}^n$ with each fibre diffeomorphic to the $(n-1)$-dimensional ball $\mathbb{B}^{n-1}$. Now, Theorem 2 implies

$$\tau(B_c) \geqslant \frac{c}{c + 2\pi}.$$

As a consequence of Corollary 3, the set B_c is contactly non-displaceable for $c > 0$. This claim, however, is not new. In fact, [18, Example 2.11] asserts that B_c is non-displaceable for $c \geqslant 0$.

[7] The loop ψ_t represents an even power of the generator of $\pi_1\big(U\big(\frac{n+1}{2}\big)\big) \cong \mathbb{Z}$ because $\frac{n+1}{2}$ is even. Therefore, ψ_t represents the trivial element in $\pi_1(O(n+1)) \cong \mathbb{Z}_2$. Consequently, ψ_t is contractible as a loop of diffeomorphisms.

1. Proof of Theorem 1

We first recall the definition of the selective symplectic homology $SH^{\Omega}_{*}(W)$. For an open subset $\Omega \subset \partial W$, denote by $\mathcal{H}_{\Omega}$ the set of the smooth autonomous contact Hamiltonians $h\colon \partial W \to [0,+\infty)$ such that[8]

(1) $\mathrm{supp} h \subset \Omega$,
(2) h vanishes up to order 2 at each $p \in \partial W$ such that $h(p) = 0$,
(3) the 1-periodic orbits of h are constant.

Denote by $\Pi(h)$ the set of the positive smooth functions $f\colon \partial W \to \mathbb{R}_{>0}$ such that the contact Hamiltonian $h+f$ has no 1-periodic orbits. We see $\mathcal{H}_{\Omega}$ and $\Pi(h)$ as directed sets with the preorder relation pointwise $\leqslant$. The selective symplectic homology is defined by

$$SH^{\Omega}_{*}(W) := \varinjlim_{h\in\mathcal{H}_{\Omega}} \varprojlim_{f\in\Pi(h)} HF_{*}(h+f), \tag{1.1}$$

where the limits are taken with respect to the continuation maps [12, Section 4]. Here, $HF_{*}(h)$ stands for the Floer homology for the contact Hamiltonian h, that was introduced in [12]. Notice that the inverse limit in (1.1) stabilizes [19, Lemma 3.5]. In the following lemma and in the rest of the paper, φ^{f}_{t} denotes the contact isotopy furnished by the contact Hamiltonian $f\colon \partial W \times \mathbb{R} \to \mathbb{R}$. The contact Hamiltonian of the composition $\varphi^{h}_{t} \circ \varphi^{f}_{t}$ is denoted by $h\#f$. Explicitly,

$$(h\#f)_{t}(x) := h_{t}(x) + \left(\kappa^{h}_{t} \cdot f_{t}\right) \circ \left(\varphi^{h}_{t}\right)^{-1}(x),$$

where κ^{h}_{t} is defined by $\left(\varphi^{h}_{t}\right)^{*}\alpha = \kappa^{h}_{t} \cdot \alpha$.

Lemma 5. *In the situation of Theorem* 1*, for every $\ell \in \mathbb{R}$ there exists $h \in \mathcal{H}_{\Omega}(\partial W)$ and a contractible loop $\varphi^{f}_{t}\colon \partial W \to \partial W$ of contactomorphisms*[9] *such that*

$$(h+\varepsilon)\#f \geqslant \ell$$

for all $\varepsilon \geqslant 0$.

Proof. Without loss of generality, assume $\ell > 0$. By the assumptions in Theorem 1, the set A is the preimage of a closed subset under a smooth Reeb-invariant map $\partial W \to \mathbb{R}$. Since any closed subset of $\mathbb{R}$ can be seen as the preimage of $\{0\}$ under some smooth function $\mathbb{R} \to \mathbb{R}$, there exists a smooth Reeb-invariant function[10] $\mu\colon \partial W \to \mathbb{R}$ such that $A = \mu^{-1}(\{0\})$. Since A is compact and immaterial, there exists (see [19, Lemma 7.2]) a contractible loop of contactomorphisms $\varphi^{f}_{t}\colon \partial W \to \partial W$ such that

$$\min_{x\in A, t\in\mathbb{R}} f_{t}(x) \geqslant 2\ell. \tag{1.2}$$

The set

$$S := \left\{x \in \partial W \mid \left(\min_{t\in\mathbb{R}} f_{t}\right)(x) \leqslant \ell\right\}$$

is closed because the function

$$\min_{t\in\mathbb{R}} f_{t} = \min_{t\in[0,1]} f_{t}\colon\ \partial W \to \mathbb{R}$$

is continuous. In addition, S and A are disjoint due to (1.2). Hence, there exists $a > 0$ such that $\mu^{-1}(-a,a)$ is disjoint from S. Let $m := \min f$ be the (global) minimum of f and let $\beta\colon \mathbb{R} \to [0,+\infty)$ be a smooth function such that

[8] The second condition is slightly different from that of [19, Definition 3.1]. However, using either of these two conditions yields the same definition of selective symplectic homology.

[9] Here and in the rest of the paper, loops $\varphi_{t}\colon \partial W \to \partial W$ of contactomorphisms are seen as 1-periodic smooth $\mathbb{R}$-families $\{\varphi_{t}\}_{t\in\mathbb{R}}$ of contactomorphisms.

[10] Obtained as the composition.

(1) there exists $\delta > 0$ such that $\beta(s) = 0$ if, and only if, $s \in [-\delta, \delta]$, and
(2) $\beta(s) \geqslant \ell - m$ for $s \notin (-a, a)$.

Consequently, $\beta \circ \mu(x) \geqslant \ell - m$ for x such that $\min_{t\in\mathbb{R}} f_t(x) \leqslant \ell$ and $\mathrm{supp}\beta \circ \mu \subset \Omega = \partial W \setminus A$. Denote $h := \beta \circ \mu$. By the construction of h, we have $h + f_t \geqslant \ell$. Therefore,

$$((h+\varepsilon)\#f)_t(x) = h(x) + \varepsilon + f_t \circ \left(\varphi_t^{h+\varepsilon}\right)^{-1}(x) = (h + \varepsilon + f_t) \circ \left(\varphi_t^{h+\varepsilon}\right)^{-1}(x) \geqslant \ell.$$

Here, we used $(h+\varepsilon) \circ \left(\varphi_t^{h+\varepsilon}\right)^{-1} = h + \varepsilon$, which holds because $h + \varepsilon$ is time-independent and Reeb-invariant. By perturbing h slightly (within the class of non-negative smooth functions having the same support as h) if necessary, one can achieve $h \in \mathcal{H}_\Omega$. This completes the proof. □

A contact Hamiltonian h is called *admissible* if its time-1 map φ_1^h has no fixed points x such that $\left(\left(\varphi_1^h\right)^*\alpha\right)\big|_x = \alpha|_x$. The admissibility condition is required for $HF_*(h)$ to be well defined. In the special case of a constant contact Hamiltonian $h \equiv a \in \mathbb{R}$, the admissibility simply means that a is not a period of a Reeb orbit. The following lemma asserts that the images of $HF_*(h)$ and $HF_*(g)$ under the canonical morphisms coincide provided h and g give rise to the same element in $\widetilde{\mathrm{Cont}}_0(\partial W)$.

Lemma 6. *Let $h_t \colon \partial W \to \mathbb{R}$ be an admissible contact Hamiltonian and let $\varphi_t^f : \partial W \to \partial W$ be a contractible loop of contactomorphisms. Then, the images of the following natural morphisms are equal:*

$$HF_*(h) \to SH_*(W),$$
$$HF_*(h\#f) \to SH_*(W).$$

Proof. Since f gives rise to a contractible loop of contactomorphisms, there exists a *zig-zag isomorphism* $HF_*(h) \to HF_*(h\#f)$ (see [19, Section 9] or [4, Section 2.1.3]). Proposition 2.4 from [4] asserts that the following diagram, consisting of the canonical morphisms and a zig-zag isomorphism, commutes:

$$\begin{array}{ccc} & SH_*(W) & \\ \nearrow & & \nwarrow \\ HF_*(h) & \xrightarrow{\text{zig-zag}} & HF_*(h\#f). \end{array}$$

From this commutative diagram, the claim of the lemma follows directly. □

Proof of Theorem 1. For an admissible contact Hamiltonian $h_t \colon \partial W \to \mathbb{R}$, denote by $\mathfrak{Im}(h)$ the image of the natural morphism $HF_*(h) \to SH_*(W)$. Let $\theta \in SH_*(W)$. Then, there exists a real number $\ell \in \mathbb{R}$ that is not a period of a Reeb orbit on ∂W such that $\theta \in \mathfrak{Im}(\ell)$. By Lemma 5, there exist $h \in \mathcal{H}_\Omega(\partial W)$ and a contractible loop $\varphi_t^f : \partial W \to \partial W$ of contactomorphisms such that the natural morphism $HF_*(\ell) \to SH_*(W)$ factors through $HF_*((h+\varepsilon)\#f) \to SH_*(W)$ for all positive $\varepsilon \in \Pi(h)$. Hence, $\mathfrak{Im}(\ell) \subset \mathfrak{Im}((h+\varepsilon)\#f)$ for all positive $\varepsilon \in \Pi(h)$. Since f generates a contractible loop of contactomorphisms, Lemma 6 implies $\mathfrak{Im}(\ell) \subset \mathfrak{Im}(h+\varepsilon)$ for all positive $\varepsilon \in \Pi(h)$. This further implies that θ belongs to the image of the continuation map $SH_*^\Omega(W) \to SH_*(W)$. Since θ was arbitrarily chosen, this completes the proof. □

2. Proof of Theorem 2

The quasi-measure τ is constructed in [4] using contact spectral invariants $c(h, \theta)$ introduced independently and simultaneously in [1] and [3].[11] The contact spectral

[11] In the present paper, we follow the approach and the conventions from [4]

invariant $c(h,\theta)$ is assigned to an element $\theta \in SH_*(W)$ and a time-dependent contact Hamiltonian $h_t\colon \partial W \to \mathbb{R}$. For the construction of τ, the case where θ is equal to the unit $e \in SH_*(W)$ is relevant. Now, we recall the properties of the contact spectral invariants that are needed in the present paper:

(1) $c(s,e) = s$ for all $s \in \mathbb{R}$.
(2) (Invariance) If $\{\varphi_t^h\}$ and $\{\varphi_t^g\}$ represent the same class in $\widetilde{\mathrm{Cont}}_0(\partial W)$, then $c(h,e) = c(g,e)$.
(3) (Monotonicity) If $h \leqslant g$ pointwise, then $c(h,e) \leqslant c(g,e)$.
(4) (Triangle inequality) If h and g are Reeb invariant, then[12]
$$c(h,e) + c(g,e) \leqslant c(h\#g, e).$$
(5) (Stability) If h and g are Reeb-invariant, then
$$\min(h-g) \leqslant c(h,e) - c(g,e) \leqslant \max(h-g).$$

For a Reeb-invariant contact Hamiltonian $h\colon \partial W \to \mathbb{R}$, the contact quasi-state $\zeta(h)$ is defined by
$$\zeta(h) := \lim_{k\to+\infty} \frac{c(k\cdot h, e)}{k}.$$
The limit exists and is finite [4, Lemma 7.2]. The contact quasi-state ζ satisfies the following homogeneity property: $\zeta(sh) = s\zeta(h)$ for $s \geqslant 0$. Now, the contact quasi-measure τ is defined by
$$\tau(A) := \inf\{\zeta(h) \mid h\colon \partial W \to [0,1] \text{ Reeb-invariant, smooth, and } h|_A = 1\}$$
for a closed subset $A \subset \partial W$. The following lemma is used in the proof of Theorem 2.

Lemma 7. *Let M be a closed contact manifold with a contact form α and let $B \subset M$ be a closed subset. Let $f_t\colon M \to \mathbb{R}$ be a contact Hamiltonian and $h\colon M \to [0,1]$ a Reeb-invariant contact Hamiltonian such that $h|_B = 1$. Denote*
$$\ell := \inf_{t\in\mathbb{R}, x\in M\setminus B} f_t(x) \qquad \text{and} \qquad m := \inf_{t\in\mathbb{R}, x\in M} f_t(x),$$
and assume m is finite.[13] Then, the following inequality holds:
$$((\ell - m)\cdot h)\#f \geqslant \ell.$$

Proof. Denote $g := (\ell - m)\cdot h$. Since g is time-independent and Reeb-invariant, it is constant along its trajectories. Consequently,
$$(g\#f)_t(x) = g(x) + f_t \circ (\varphi_t^g)^{-1}(x) = (g + f_t)\circ(\varphi_t^g)^{-1}(x).$$
Therefore, it is enough to prove $g + f_t \geqslant \ell$. If $x \in M \setminus B$, then $f_t(x) \geqslant \ell$. Since g is non-negative, this implies $g(x) + f_t(x) \geqslant \ell$. If, on the other hand, $x \notin M \setminus B$, then $x \in B$ and
$$g(x) + f_t(x) = \ell - m + f_t(x) \geqslant \ell.$$
This finishes the proof. □

Proof of Theorem 2. Let $h\colon \partial W \to [0,1]$ be a Reeb-invariant contact Hamiltonian such that $h|_B = 1$. By Lemma 7, we get $((m_B - m)\cdot h)\#f \geqslant m_B$. The invariance and the monotonicity properties of the contact spectral invariants imply
$$c((m_B - m)\cdot h, e) = c(((m_B - m)\cdot h)\#f, e) \geqslant c(m_B, e) = m_B.$$
The same argument applied to the iterated loop φ_{kt}^f (generated by the contact Hamiltonian $k\cdot f_{kt}$), $k \in \mathbb{N}$, implies
$$c(k\cdot(m_B - m)\cdot h, e) \geqslant k\cdot m_B.$$

[12]Notice that, due to our conventions, the triangle inequality is "reversed".
[13]This implies ℓ is finite as well.

Therefore,

$$\zeta((m_B - m) \cdot h) = \lim_{k \to +\infty} \frac{c(k \cdot (m_B - m) \cdot h, e)}{k} \geqslant m_B.$$

Consequently, due to homogeneity of ζ, we have $\zeta(h) \geqslant \frac{m_B}{m_B - m}$. This completes the proof. □

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